

Unit

2

Lesson

1



Innovators in Action.

Up & Down for Everyone!

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Design a seesaw that allows children who use wheelchairs to play safely.
- Explore and describe how force and motion affect how the seesaw moves.
- Use addition and subtraction up to 20 to balance different weights on each side.
- Measure and compare objects to help create a balanced design.
- Choose strong and safe materials for building their seesaw.
- Observe how pushes and pulls make the seesaw move.
- Use words like fast, slow, up, and down to describe movement.

Challenge questions of the lesson

1. How can you make it easy and safe for a friend in a wheelchair to get on and off the seesaw by themselves?
2. If the seesaw needs to be strong, what things could we use to build it so it doesn't break when we play?
3. How can we make sure the seesaw has enough space so lots of friends can play on it at the same time, even with a wheelchair?

SEL Relation

- Empathy and Inclusive Design (Social Awareness): Students will deeply consider the needs and experiences of peers with disabilities, specifically wheelchair users, throughout their design process, fostering empathy and understanding.
- Collaboration and Communication (Relationship Skills): Students will work together (if in groups) to brainstorm ideas, share design concepts, and provide constructive feedback, developing their communication and teamwork skills.
- Problem-Solving and Innovation (Self-Awareness & Responsible Decision-Making): Students will face the complex challenge of creating a truly accessible and functional seesaw, requiring them to think creatively, analyze constraints, and make responsible design choices.
- Promoting Equity and Fairness (Social Awareness & Responsible Decision-Making): The lesson inherently encourages students to think about creating equitable play opportunities, fostering a sense of fairness and social responsibility within the classroom community.

Engage



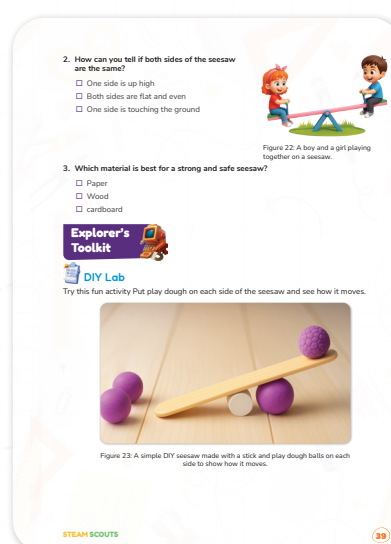
Teacher Role:

- **Facilitator:** Guide the discussion about the story, ensuring all students understand the problem with the seesaw and the initial ideas for making it better.
- **Questioner:** Pose open-ended questions to encourage students to think about how seesaws work, the concept of balance, and how to make playground equipment accessible.
- **Connector:** Help students connect the characters' ideas (changing the middle, changing how it moves, creating safe space) to potential design solutions.
- **Motivator:** Build enthusiasm for the upcoming activities focused on exploring and redesigning the seesaw.
- **Inclusion Advocate:** Reinforce the importance of making playground equipment usable and enjoyable for all children.

Expected From Students:

- **Active Listening:** Students will listen attentively to the story and identify the problem Adam and Khaled raise about the seesaw.
- **Empathy:** Students will consider Khaled's perspective and understand the desire to make the seesaw easier for him to use.
- **Idea Generation:** Students will begin to brainstorm ways to modify a seesaw to make it easier to move and safer for everyone.
- **Participation in Discussion:** Students will share their initial thoughts and reactions to the characters' ideas and contribute their own suggestions.
- **Problem-Solving (Initial):** Students will start thinking about the basic mechanics of a seesaw and potential ways to overcome the challenges mentioned.

Parts of the books included.



Story



- Hook with a Real-World Problem

Begin by saying:

“Have you ever been on a seesaw where one person is much heavier than the other? What happens?”

Let students respond.

Then add:

“What if someone uses a wheelchair—how could they ride a seesaw too?”

- Introduce the Story

Say:

“Today, we’re going to listen to a story where Adam, Laila, and Khaled face this challenge. Let’s find out how they start solving it.”

Read the scene aloud using expressive voices for each character. Highlight the feelings in Khaled’s voice and the problem-solving spirit in Adam and Laila’s.

- Make the Problem Personal

After reading, ask:

“What do you think would help Khaled enjoy the seesaw?”

“What makes a seesaw fun AND fair for everyone?”

- Bridge to Design Challenge

Say:

“Just like the kids in the story, we’re going to become engineers today. We’ll think of ideas, plan, and build a new kind of seesaw—one that works for everyone.”

Possible Strategies for Implementation:

- Think-Pair-Share

Let students think about Khaled’s challenge, pair up, and share one idea they have for making the seesaw more fair.

- Draw the Problem

Have students draw the seesaw as it is now and how they would change it to make it more inclusive.

- Role-Play Engineering Team

Create teams where students take on roles: designer, safety checker, and tester. Each group can talk through the ideas just like the characters in the story.

- Story Extension

Ask students to write or draw what might happen next in the story after the class begins to build.

- Emotion Check-In

Prompt students with:

“How do you think Khaled felt when he couldn’t use the seesaw?”

“How do you think he will feel when it’s fixed?”

This supports empathy-based design thinking.

Let's Think!



Objective: Students will develop their understanding of how motion and force affect the movement of a seesaw, how to identify balance and symmetry in design, and which materials are appropriate for building strong and safe structures. This activity promotes scientific reasoning, observation, and application of engineering principles in a playful, accessible way.

Possible Answers:

1. Which of these makes the seesaw move faster?

- ☐ A gentle push/pull
- ☒ A big push/pull (More force generally leads to faster movement)
- ☐ No push/pull (Without force, the seesaw won't move)

2. How can you tell if both sides of the seesaw are the same (balanced)?

- ☐ One side is up high (Indicates an imbalance)
- ☒ Both sides are flat and even (Suggests equal weight or no force being applied)
- ☐ One side is touching the ground (Indicates an imbalance)

3. Which material is best for a strong and safe seesaw?

- ☐ Paper (Weak and easily broken)
- ☒ Wood (Generally strong and durable for outdoor equipment)
- ☐ Cardboard (Not strong enough for repeated use and weight)

Possible Strategies for implementation:

- Read-Aloud and Discuss:

Read each question aloud and allow students to respond orally before marking their choices. This supports early readers and English language learners.

- Demonstration with Mini Seesaw Model:

Use a simple ruler-and-pencil seesaw model to physically show how big pushes make it move, and what “flat and even” looks like.

- Think-Pair-Share:

Have students discuss their answers in pairs before marking them. For example, “What happens if we push hard on the seesaw?”

Explorer's Toolkit



Teacher Role:

- **Introducer:** Introduce the activity as a way to explore how weight affects the movement of a seesaw. Explain that they will use playdough to represent weight and observe how it makes the seesaw move.
- **Facilitator:** Guide students through the steps of placing playdough on the seesaw and observing its movement. Ensure they understand how to record their observations in their CER journals.
- **Questioner:** Prompt students to think about the relationship between the amount of playdough (weight) and the movement of the seesaw. Ask guiding questions to help them connect their observations to the concepts of balance and force.
- **Observer:** Observe students as they conduct the experiment and record their findings, noting their understanding of cause and effect.

Expected From Students:

- **Hands-on Participation:** Students will actively participate in placing playdough on the seesaw and observing its movement.
- **Following Instructions:** Students will follow the step-by-step instructions to conduct the experiment.
- **Observation:** Students will carefully watch how the seesaw moves when different amounts of playdough are placed on each side.
- **Data Recording:** Students will record their observations in their CER (Claim, Evidence, Reasoning) journals, following the prompts provided.
- **Developing Understanding of Balance:** Students will begin to understand how the distribution of weight affects the balance and movement of a seesaw.

Parts of the books included.





DIY Lab

Objective: Students will explore the relationship between weight and movement by testing how adding more marbles to one side of a seesaw affects its motion. They will record their observations using the CER model to support scientific thinking and communication.

Materials Needed:

- Small seesaw (as shown in the image)
- Playdough (in small, roughly equal amounts or easily divisible)
- CER Journal (or a worksheet with the CER chart provided)



Activity Instructions:

1. Prepare Materials in Advance:

- Set up simple seesaw models using a flat cardboard strip balanced on a pencil, ruler, or round stick.
- Place small cups or containers of marbles at each group station (at least 10 marbles per group).
- Print and distribute the CER Journal worksheet (Claim–Evidence–Reasoning).

2. Introduce the Activity:

- Gather students and say:
“Today we’re going to investigate how marbles affect the way a seesaw moves. You’ll be real scientists!”
- Ask:
“What happens if one side of the seesaw has more marbles than the other?”
“What do you think makes it tip faster or slower?”

3. Demonstrate the Setup:

- Use a sample seesaw to show how placing more marbles on one side causes it to tilt.
- Emphasize what to observe: movement direction, speed, and balance.

4. Guide Students Through Their CER Journal:

- Briefly explain the 3 parts:
 - Claim – What do you think will happen when you add more marbles?
 - Evidence – What did you see? How many marbles made it tip?
 - Reasoning – Why did the seesaw move the way it did?
- Model one example aloud as a class.

5. Hands-On Exploration:

- Students work in pairs or small groups to:

- Add marbles to each side of their seesaw.
- Observe the movement.
- Fill out their CER Journal with their findings.

6. Support Students as They Work:

- Move around the room asking prompts:
“How many marbles did it take to make it tip?”
“What happened when you added more?”
“Why did it move that way?”

7. Conclude with Class Reflection:

- Have students share their favorite discoveries.
- Reinforce that heavier weight causes one side to go down and balance means equal weight on both sides.



- ☐ Faster
- ☐ Slower
- ☐ Stays the same

Possible Strategies for Implementation:

- **Hands-On Small Group Testing**
Let students work in pairs or small groups to test with different amounts of marbles/playdough. This promotes teamwork and observation.
- **Guided CER Modeling**
Model the first part (Claim) as a class, then let students try the Evidence and Reasoning parts individually or with a partner.
- **Use Visual Cues**
Post a poster or slide showing what Claim–Evidence–Reasoning means in simple terms to support young learners.
- **Think Aloud**
Ask guiding questions like:
“What did you see when we added more to one side?”
“Which side moved first? Why do you think that happened?”
- **Use Real-Life Connections**
Compare to real playground seesaws and talk about how weight affects movement—helping students link classroom experiments to real-world play.

Explain



Teacher Role:

- **Concept Introducer:** Explain that the video will demonstrate how pushes and pulls (forces) make things move, specifically using the example of a seesaw. Highlight that they will learn about how the direction and strength of a push or pull affect the seesaw's movement.
- **Observation Guider:** Direct students' attention to key aspects in the video, such as the direction of the push or pull, the resulting movement of the seesaw, and how different weights might affect this movement.
- **Discussion Facilitator:** Lead a post-viewing discussion to help students articulate their observations and connect them to the concepts of force and motion. Clarify any misconceptions.
- **Vocabulary Builder:** Introduce and reinforce terms like "force," "push," "pull," "up," "down," and "heavy."

Expected From Students:

- **Focused Viewing:** Students will watch the video attentively to understand how pushes and pulls cause the seesaw to move.
- **Observation Identification:** Students will identify and note the direction of the forces applied in the video and the corresponding movement of the seesaw.

- **Active Thinking:** Students will consider how the ideas in the video relate to their own experiences with seesaws and other moving objects.
- **Verbal Sharing:** Students will participate in the pre- and post-watching discussions, sharing their initial thoughts and observations from the video.

Parts of the books included.



Screen Lessons

Let's click the link and explore how things move with pushes and pulls—let's learn something new together!

https://www.youtube.com/watch?v=Se_e5h9vtms



Screen Lessons

Objectives: Students will observe how varying weights and their positions on a seesaw affect its balance and movement. This visual demonstration will reinforce concepts of force, motion, and equilibrium, providing a foundation for designing inclusive playground equipment that accommodates children of all abilities.

Video link: https://www.youtube.com/watch?v=Se_e5h9vtms



Possible Strategies for implementation:

1. Pre-Watching: Set a purpose.

- Strategy: "Personal Connection": Begin by asking students to share their own experiences with seesaws. Ask questions like: "What do you do to make a seesaw go up?" "What do you do to make it go down?" "What happens if someone pushes or pulls really hard?" This connects the video to their prior knowledge.

2. During Watching: Active engagement.

- Strategy: "Motion Tracker": Provide students with a simple chart or ask them to mentally track what happens when someone pushes down on one side of the seesaw and what happens to the other side. Encourage them to also note if the video shows different strengths of pushes or pulls and the resulting speed of movement.

3. Post-Watching: Discussion and reflection.

- Strategy: "Cause and Effect": After the video, ask students direct questions about cause and effect: "What was the push or pull (the cause) and what happened to the seesaw (the effect)?" Encourage them to use the terms "force," "push," and "pull" in their answers.

Elaborate



Teacher Role:

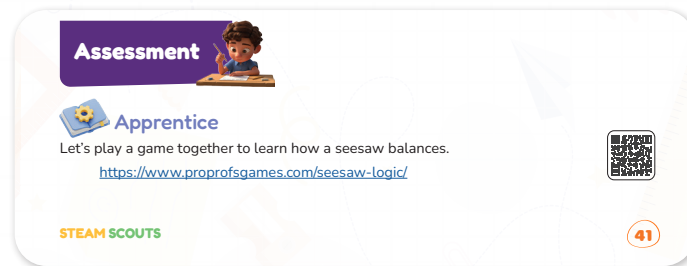
- **Facilitator:** Guide students in accessing and playing the "Seesaw Logic" game.
- **Encourager:** Encourage students to explore different strategies for balancing the seesaw in the game.

- **Observer:** Observe students' problem-solving approaches as they play the game.

Expected From Students:

- **Game Engagement:** Students will actively play the "Seesaw Logic" game.
- **Exploration:** Students will experiment with placing weights on different parts of the seesaw to achieve balance.
- **Understanding Balance:** Students will begin to understand the concept of balance on a seesaw through gameplay.

Parts of the books included.



Apprentice

Objective:

Students will develop logical thinking and problem-solving skills by experimenting with digital seesaw balance challenges. Through this activity, they will explore weight distribution and equilibrium, applying physics concepts in a playful, interactive format that supports inclusive playground design thinking.

Activity link: <https://www.proprofsgames.com/seesaw-logic/>



Activity Instructions:

- Introduce the Game
 - Say: "Today you'll be using your thinking and problem-solving skills to balance a seesaw using animals of different sizes and weights. Just like in real life, things need to be fair and balanced so everyone can play safely."
- Access the Game
 - Direct students to the game link.
 - Ensure devices are charged and internet-connected.
- Model First Round (Optional)
 - Project the game on a screen and play through the first round together. Discuss your thinking aloud (e.g., "The elephant is heavier, so I need more small animals on the other side to balance it.").
- Independent or Partner Play
 - Let students play the game alone or in pairs. Encourage verbal reasoning: "Why did you choose that animal?"
 - Allow time for replay and experimentation.

- Reflect as a Group

Discuss questions like:

- “Which levels were tricky? Why?”
- “How is this like a real seesaw?”
- “How can we use this learning in our playground designs?”

Possible Strategies for Implementation:

- **Think-Aloud Modeling:** Teachers demonstrate one or two levels while talking through their logic.
- **Pair and Share:** Students work in pairs, discussing and negotiating weight placements.
- **Supportive Grouping:** Pair stronger readers with peers who might need help navigating the interface or instructions.

Evaluate



Teacher Role:

- **Facilitator:** Organize a "Seesaw Showcase" where students present their designs for a strong, wide, and easy-to-use seesaw with wheelchair space.
- **Observer:** Observe students as they present, noting how they have incorporated features for strength, width, ease of use, and wheelchair accessibility.
- **Questioner:** Ask probing questions to encourage students to explain their design choices and how they address the needs of all users, particularly those in wheelchairs.
- **Feedback Provider:** Offer constructive feedback based on the rubric, highlighting strengths and areas for potential improvement in their design thinking.
- **Connector:** Help students link their design ideas back to the lessons on forces, balance, materials, and inclusive playground design.

Expected From Students:

- **Project Completion:** Students will have developed a design (drawing, model, or detailed description) for an accessible seesaw.
- **Presentation:** Students will present their design, explaining its features for strength, width, ease of use, and wheelchair accessibility.
- **Demonstration of Understanding:** Students will demonstrate an understanding of the principles of balance, material strength, and inclusive design.
- **Articulation of Design Choices:** Students will be able to explain why they made specific design decisions to accommodate wheelchair users and ensure safety and ease of use for everyone.

Parts of the books included



Figure 25: A wide and strong seesaw designed with space for a wheelchair so all friends can play together safely.



Showcase

Objective: Students will design and build a model of an inclusive seesaw that is strong, wide, and safe, with space to support children using wheelchairs. This activity helps students apply engineering, design, and empathy skills by constructing a hands-on model that reflects accessibility and fairness in play.

Materials Needed:

- Craft sticks, large popsicle sticks, or wooden rulers (for the seesaw board)
- Toilet paper tubes, blocks, or foam pieces (for the fulcrum/base)
- Strong cardboard for platform and wheelchair space
- Scissors and glue/hot glue
- Ruler for measuring width
- Markers or labels for decoration
- Optional: toy wheelchair models, small figures, marbles for testing movement

Activity Instructions:

1. Review the Goal:

- Start with: “We’re building a seesaw that works for everyone—including friends who use wheelchairs.”
- Show a photo/model of a wheelchair-accessible seesaw and ask:

“What features make this easier to use?”

2. Plan the Design:

- Ask students to sketch their design on paper first.
- Discuss width, ramp space, and balance. Let them label where the wheelchair area will go.

3. Build the Model:

- Guide students to construct their seesaw using the materials:
 - Use a wide base and ensure stability.

- Include a flat platform with side rails or wide seating space.
- Create ramps (e.g., folded cardboard) on either side.

4. Test and Improve:

- Place weights (marbles or toy figures) on both sides.
- See how the seesaw moves.
- Ask: “Does it balance? Is it easy to use?”

5. Decorate and Display:

- Let students personalize their seesaws and explain how it helps all children play.



Possible Strategies for Implementation:

- **Use a Design Checklist:** Ensure students include safety rails, wide platforms, and space for wheelchairs.
- **Inclusion Lens:** Have students reflect: “How would Khaled use this seesaw?”
- **Peer Feedback:** Pair students to test and give ideas for improvement.

Showcase: Rubric for Evaluating Accessible Seesaw Designs

Criteria	Beginning (1 Point)	Developing (2 Points)	Proficient (3 Points)
Strength & Durability	Design shows limited consideration for the seesaw's strength. Materials or construction methods are unclear or appear weak.	Design indicates some awareness of the need for a strong seesaw, but details about materials or construction are basic.	Design clearly considers the strength and durability of the seesaw, with potential materials (e.g., strong wood, metal) or structural elements suggested.
Width & Ease of Use	Design does not appear wide enough for comfortable use by multiple children or shows features that might make it difficult to use.	Design shows some consideration for width and ease of use, but some aspects might still be narrow or challenging for some users.	Design demonstrates a good understanding of width for multiple users and incorporates features for ease of getting on and off and comfortable use for all.
Wheelchair Space	Design does not include dedicated space for a wheelchair or shows a very limited or unsafe accommodation.	Design includes some designated space for a wheelchair, but the accessibility or safety of this space may be unclear or potentially difficult.	Design clearly provides a safe and accessible space for a wheelchair to be integrated into the seesaw, allowing the wheelchair user to participate alongside others.
Safety Considerations	Design shows limited awareness of safety features for all users.	Design includes some basic safety considerations, but they may not be comprehensive or specific to wheelchair users.	Design demonstrates thoughtful consideration for the safety of all users, including securement for wheelchairs (if applicable), handholds, and potentially controlled movement.
Presentation & Explanation	Student struggles to explain their design or how it addresses strength, width, ease of use, and wheelchair accessibility.	Student provides a basic explanation of their design but may not clearly articulate all the features for accessibility and usability.	Student clearly explains their design choices, detailing how the seesaw is strong, wide, easy to use, and specifically accommodates a wheelchair user safely and inclusively.

It's time to move to the create part and add this.



Now I Can

Objective: Students will reflect on their learning by identifying key science, math, and design concepts used while building a seesaw. They will demonstrate understanding of force and motion, measurement, and inclusive design by describing how their seesaw works, what they used to build it, and how it helps all children play together safely.

Activity Instructions

- Review Each “Now I Can” Statement Together:
Read each statement aloud and ask for examples from the activity to reinforce meaning (e.g., “What materials did you pick to make your seesaw strong?”).
- Encourage Self-Assessment:
Give students a copy of the “Now I Can” checklist. Ask them to check off each skill they used during the seesaw project.

Relation to Standards:

1. Next Generation Science Standards (NGSS):
 - **K-2-ETS1-2:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem. (Students are designing a functional and accessible piece of equipment.)
 - **K-2-ETS1-3:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs. (Implicitly, students are designing a solution to the problem of inaccessible seesaws.)
2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.1.4:** Describe people, places, things, and events with relevant details, expressing ideas and feelings clearly. (Students present and explain their seesaw designs.)
- **SL.1.5:** Add drawings or other visual displays to descriptions when appropriate to clarify ideas, thoughts, and feelings. (The design plan serves as a visual display.)

3. Common Core State Standards (CCSS) - Mathematics (Geometry):

- **1.G.A.3:** Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles, and quarter-circles) to create a composite shape, and compose new shapes from the composite shapes. (Students are likely using geometric shapes in their designs.)